University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
3. Write your answers in the spaces provided. If you must use more room, use the space available on the last three pages, and indicate this on the question page where you ran out of room.
4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$
$$q_0 \ \sqcup \ q_1 \ 1 \ R$$
$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$
$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a *k-clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

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6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$
$$q_0 \; ␣ \; q_1 \; 1 \; R$$
$$q_1 \; ␣ \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(␣) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; ␣ \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \sqcup \; q_1 \; 1 \; R$$

$$q_1 \; \sqcup \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \sqcup q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x \colon x \in A\} \cup \{3x+1 \colon x \in B\} \cup \{3x+2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$

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(a) (3 points)
Complete the following table to describe a possible transition function δ:

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$$f(x) = \_\_\_\_\_\_\_\_$$

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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Student ID number

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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \text{␣}\ q_1\ 1\ R$$
$$q_1\ \text{␣}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{␣}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) $\mathrm{P} \subseteq \mathrm{NP}$.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

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Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

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(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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University of Toronto-Mississauga

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(a) (3 points)
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	1	$\llcorner\!\lrcorner$
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Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

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Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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CSC363 Final Examination

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$$f(x) = \_\_\_\_\_\_\_\_$$

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a *k-clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$
$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$
$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

$$q_0\ \text{\textvisiblespace}\ q_1\ 1\ R$$

$$q_1\ \text{\textvisiblespace}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\text{\textvisiblespace}$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{\textvisiblespace}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{\textvisiblespace}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

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Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

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For every partial recursive function g, the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

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(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \text{␣}\ q_1\ 1\ R$$
$$q_1\ \text{␣}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{␣}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

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(c) A set is recognizable if and only if it is c.e. (computably enumerable).

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Q6. Answer each of the following

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Let L be the following language:

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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Let L be the following language:

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
3. Write your answers in the spaces provided. If you must use more room, use the space available on the last three pages, and indicate this on the question page where you ran out of room.
4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$
$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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$$q_0\ \sqcup\ q_1\ 1\ R$$

$$q_1\ \sqcup\ q_h\ 1\ R$$

(a) (3 points)
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q_1		

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Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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(a) (3 points)
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Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x+1 \colon x \in B\} \cup \{3x+2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$

$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N}\colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

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$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

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2. Every two vertices in C are connected by an edge in the graph.

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

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CSC363 Final Examination

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CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
3. Write your answers in the spaces provided. If you must use more room, use the space available on the last three pages, and indicate this on the question page where you ran out of room.
4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$
$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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$$q_1\ \text{␣}\ q_h\ 1\ R$$

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Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

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Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Q4. Write True or False with a brief justification (2 points each)

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(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that 3SAT $\leq_p L$.

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(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \text{␣} \; q_1 \; 1 \; R$$

$$q_1 \; \text{␣} \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \text{␣} \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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Q9. (2 points)

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a *k-clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$
$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
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	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

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(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
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Q6. Answer each of the following

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Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \sqcup\ q_1\ 1\ R$$
$$q_1\ \sqcup\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \sqcup\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

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Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$
$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$
$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N}\colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

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Prove that 3SAT $\leq_p L$.

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(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$

$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; ␣ \; q_1 \; 1 \; R$$

$$q_1 \; ␣ \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(␣) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; ␣ \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

$$q_0\ \textvisiblespace\ q_1\ 1\ R$$

$$q_1\ \textvisiblespace\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\textvisiblespace$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\textvisiblespace) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \textvisiblespace\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N}\colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

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(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

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For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
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Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x + 1\colon x \in B\} \cup \{3x + 2\colon x \in C\}$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Q8. (5 points)

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a *k-clique* if there exists $C \subseteq V$ such that:

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Let L be the following language:

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
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6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$

$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x+1 \colon x \in B\} \cup \{3x+2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that $3\text{SAT} \leq_p L$.

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that 3SAT $\leq_p L$.

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Q4. Write True or False with a brief justification (2 points each)

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Q4. Write True or False with a brief justification (2 points each)

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(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$
$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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i. Prove that $D \leq_m E$.

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Q8. (5 points)

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Q9. (2 points)

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

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Last name

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
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(a) (3 points)
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	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
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Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \text{␣}\ q_1\ 1\ R$$
$$q_1\ \text{␣}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{␣}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

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(a) (3 points)
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Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \text{␣} \; q_1 \; 1 \; R$$

$$q_1 \; \text{␣} \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \text{␣} \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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$$f(x) = \_\_\_\_\_\_\_\_$$

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Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$
$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$
$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

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(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

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Q1. Consider the following Turing program:

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(a) (3 points)
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(b) (2 points)
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$$f(x) = \_\_\_\_\_\_\_\_$$

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

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Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
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(a) (3 points)
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$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

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Prove that:

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CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
3. Write your answers in the spaces provided. If you must use more room, use the space available on the last three pages, and indicate this on the question page where you ran out of room.
4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

$$q_0\ \text{␣}\ q_1\ 1\ R$$

$$q_1\ \text{␣}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{␣}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x\colon x \in A\} \cup \{3x + 1\colon x \in B\} \cup \{3x + 2\colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
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Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

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$$q_0 \; 1 \; q_0 \; 1 \; R$$

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(a) (3 points)
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(b) (2 points)
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$$f(x) = \text{--------}$$

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Q2. (3 points)

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(e) Every Boolean formula is equivalent to a 3CNF formula.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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(a) (4 points)

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Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a *k-clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

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Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$

$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

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(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
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For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
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Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

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$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
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CSC363 Final Examination

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$$q_0\ 1\ q_0\ 1\ R$$
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Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \text{␣}\ q_1\ 1\ R$$
$$q_1\ \text{␣}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{␣}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

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(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a *k-clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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Q1. Consider the following Turing program:

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(a) (3 points)
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Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$
$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
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(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

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(g) The Halting Problem is an NP-complete problem.

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CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$
$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$
$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

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Let L be the following language:

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \text{␣} \; q_1 \; 1 \; R$$

$$q_1 \; \text{␣} \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

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q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

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Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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University of Toronto-Mississauga

CSC363 Final Examination

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Q1. Consider the following Turing program:

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$$f(x) = \_\_\_\_\_\_\_\_$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Q2. (3 points)

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Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

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(g) The Halting Problem is an NP-complete problem.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that 3SAT $\leq_p L$.

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CSC363 Final Examination

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \text{␣}\ q_1\ 1\ R$$
$$q_1\ \text{␣}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{␣}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

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Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

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Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Let L be the following language:

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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University of Toronto-Mississauga

CSC363 Final Examination

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$$f(x) = \_\_\_\_\_\_\_\_$$

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
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6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$
$$q_0 \; ␣ \; q_1 \; 1 \; R$$
$$q_1 \; ␣ \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(␣) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; ␣ \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x\colon x \in A\} \cup \{3x + 1\colon x \in B\} \cup \{3x + 2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

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Let L be the following language:

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$$f(x) = \text{--------}$$

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(a) (3 points)
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CSC363 Final Examination

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Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$
$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

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(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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i. Prove that $D \leq_m E$.
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Q8. (5 points)

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Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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Last name

Student ID number

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Q1. Consider the following Turing program:

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(a) (3 points)
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$$f(x) = \text{--------}$$

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Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

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(a) Every recognizable Language is decidable.

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(g) The Halting Problem is an NP-complete problem.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

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Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

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Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
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ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

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Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
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(a) (3 points)
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	1	$\llcorner\!\lrcorner$
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$$f(x) = \_\_\_\_\_\_\_\_$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

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Prove that:

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CSC363 Final Examination

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Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a *k-clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \textvisiblespace\ q_1\ 1\ R$$
$$q_1\ \textvisiblespace\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\textvisiblespace$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\textvisiblespace) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \textvisiblespace\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \text{␣}\ q_1\ 1\ R$$
$$q_1\ \text{␣}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{␣}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
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(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that $3\text{SAT} \leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$

$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

$$q_0\ \text{␣}\ q_1\ 1\ R$$

$$q_1\ \text{␣}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{␣}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

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Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
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(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$
$$q_0 \; \sqcup \; q_1 \; 1 \; R$$
$$q_1 \; \sqcup \; q_h \; 1 \; R$$

(a) (3 points)
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	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
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$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \sqcup q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$
$$q_0 \; \sqcup \; q_1 \; 1 \; R$$
$$q_1 \; \sqcup \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

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(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

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Q1. Consider the following Turing program:

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(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

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Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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Q8. (5 points)

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Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a *k-clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

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$$q_0 \; 1 \; q_0 \; 1 \; R$$
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$$f(x) = \_\_\_\_\_\_\_\_$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

$$q_0\ \textvisiblespace\ q_1\ 1\ R$$

$$q_1\ \textvisiblespace\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\textvisiblespace) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \textvisiblespace\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) $\mathrm{P} \subseteq \mathrm{NP}$.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

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Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
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(b) (6 points)
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$$q_0 \; 1 \; q_0 \; 1 \; R$$
$$q_0 \; \text{␣} \; q_1 \; 1 \; R$$
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(a) (3 points)
Complete the following table to describe a possible transition function δ:

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Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

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(e) Every Boolean formula is equivalent to a 3CNF formula.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

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1. C has k-many elements (vertices).
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Let L be the following language:

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

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Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

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(a) (3 points)
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q_0		$q_1, 1, R$
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(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{\textvisiblespace}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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Q4. Write True or False with a brief justification (2 points each)

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(d) P $\subseteq$ NP.

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(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

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Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

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Hint: Prove the contrapositive of the statement.

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Let L be the following language:

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

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$$q_0\ 1\ q_0\ 1\ R$$
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$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \text{␣} \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
3. Write your answers in the spaces provided. If you must use more room, use the space available on the last three pages, and indicate this on the question page where you ran out of room.
4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

$$q_0\ \text{\textvisiblespace}\ q_1\ 1\ R$$

$$q_1\ \text{\textvisiblespace}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\text{\textvisiblespace}$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{\textvisiblespace}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{\textvisiblespace}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

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Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
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(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

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Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$

$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x\colon x \in A\} \cup \{3x + 1\colon x \in B\} \cup \{3x + 2\colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$
$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$
$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

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(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

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Q6. Answer each of the following

(a) (3 points)
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Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)
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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Q8. (5 points)

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Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

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Last name

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$$q_0 \; \sqcup \; q_1 \; 1 \; R$$
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(a) (3 points)
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	1	$\sqcup$
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(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

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Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$
$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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Q9. (2 points)

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

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Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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(d) P $\subseteq$ NP.

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(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

$$q_0 \ \text{␣}\ q_1\ 1\ R$$

$$q_1 \ \text{␣}\ q_h\ 1\ R$$

(a) (3 points)
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(b) (2 points)
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$$f(x) = \_\_\_\_\_\_\_\_$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

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(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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(a) (3 points)
Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

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Let L be the following language:

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$

$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

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(a) Every recognizable Language is decidable.

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(e) Every Boolean formula is equivalent to a 3CNF formula.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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Q9. (2 points)

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
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(a) (3 points)
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q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

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Q4. Write True or False with a brief justification (2 points each)

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

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Q7. Let A, B, and C be three sets of integers.
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Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

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Student ID number

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
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(a) (3 points)
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$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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Prove that:

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(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \llcorner\!\lrcorner \ q_1 \ 1 \ R$$

$$q_1 \ \llcorner\!\lrcorner \ q_h \ 1 \ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \llcorner\!\lrcorner \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \text{\textvisiblespace} \; q_1 \; 1 \; R$$

$$q_1 \; \text{\textvisiblespace} \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\text{\textvisiblespace}$
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q_1		

(b) (2 points)
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$$f(x) = \text{--------}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{\textvisiblespace}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

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(e) Every Boolean formula is equivalent to a 3CNF formula.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Q6. Answer each of the following

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Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

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(a) (3 points)
Complete the following table to describe a possible transition function δ:

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q_0		$q_1, 1, R$
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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

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(c) A set is recognizable if and only if it is c.e. (computably enumerable).

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

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Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Let L be the following language:

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

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Q2. (3 points)

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q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$
$$q_0 \ \textvisiblespace \ q_1 \ 1 \ R$$
$$q_1 \ \textvisiblespace \ q_h \ 1 \ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\textvisiblespace$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\textvisiblespace) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \textvisiblespace \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

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(c) A set is recognizable if and only if it is c.e. (computably enumerable).

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(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

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5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \text{␣}\ q_1\ 1\ R$$
$$q_1\ \text{␣}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{␣}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

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(a) (1 point)
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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$

$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
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(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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Prove that:

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Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x + 1\colon x \in B\} \cup \{3x + 2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
3. Write your answers in the spaces provided. If you must use more room, use the space available on the last three pages, and indicate this on the question page where you ran out of room.
4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$
$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \textvisiblespace\ q_1\ 1\ R$$
$$q_1\ \textvisiblespace\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\textvisiblespace$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\textvisiblespace) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \textvisiblespace\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
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(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \text{␣}\ q_1\ 1\ R$$
$$q_1\ \text{␣}\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \text{␣}\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that $3\text{SAT} \leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

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(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
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$$f(x) = \text{--------}$$

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

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(g) The Halting Problem is an NP-complete problem.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

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$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
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(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Hint: Prove the contrapositive of the statement.

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1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
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University of Toronto-Mississauga

CSC363 Final Examination

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Q1. Consider the following Turing program:

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(a) (3 points)
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q_0		$q_1, 1, R$
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(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Q6. Answer each of the following

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Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

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Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \text{␣} \; q_1 \; 1 \; R$$

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(a) (3 points)
Complete the following table to describe a possible transition function δ:

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$$f(x) = \_\_\_\_\_\_\_\_$$

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

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Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k)\colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
3. Write your answers in the spaces provided. If you must use more room, use the space available on the last three pages, and indicate this on the question page where you ran out of room.
4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$

$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$

$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

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Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

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Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
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Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \sqcup \; q_1 \; 1 \; R$$

$$q_1 \; \sqcup \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \sqcup q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

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(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Q8. (5 points)

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Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

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Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
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Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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Q1. Consider the following Turing program:

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(a) (3 points)
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$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

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(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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(a) (4 points)
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(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
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i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$
$$q_0 \; \text{␣} \; q_1 \; 1 \; R$$
$$q_1 \; \text{␣} \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
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Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

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(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that 3SAT $\leq_p L$.

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$$q_0\ 1\ q_0\ 1\ R$$

$$q_0\ \textvisiblespace\ q_1\ 1\ R$$

$$q_1\ \textvisiblespace\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\textvisiblespace$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \_\_\_\_\_\_\_\_$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\textvisiblespace) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0\ \textvisiblespace\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N}\colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

Q5. Answer each of the following

(a) (1 point)
Give an example of a set A of natural numbers which is both c.e. and co-c.e. (co-c.e. means the complement of a c.e. set).

(b) (1 point)
Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)
Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.
Let $D = \{3x\colon x \in A\} \cup \{3x+1\colon x \in B\} \cup \{3x+2\colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R$$

$$q_1 \; \llcorner\!\lrcorner \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \llcorner\!\lrcorner \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)
For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)
For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

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(a) (4 points)
Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.
ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

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Q1. Consider the following Turing program:

$$q_0\ 1\ q_0\ 1\ R$$
$$q_0\ \llcorner\!\lrcorner\ q_1\ 1\ R$$
$$q_1\ \llcorner\!\lrcorner\ q_h\ 1\ R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	$\llcorner\!\lrcorner$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{--------}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\llcorner\!\lrcorner) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \llcorner\!\lrcorner\ q_1\ 1\ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

Q4. Write True or False with a brief justification (2 points each)

(a) Every recognizable Language is decidable.

(b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

(c) A set is recognizable if and only if it is c.e. (computably enumerable).

(d) P $\subseteq$ NP.

(e) Every Boolean formula is equivalent to a 3CNF formula.

(f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

(g) The Halting Problem is an NP-complete problem.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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For A, C above: Is $A \leq_T C$? Justify your answer.

Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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University of Toronto-Mississauga

CSC363 Final Examination

Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Instructions

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5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.

Q1. Consider the following Turing program:

$$q_0 \; 1 \; q_0 \; 1 \; R$$

$$q_0 \; \text{␣} \; q_1 \; 1 \; R$$

$$q_1 \; \text{␣} \; q_h \; 1 \; R$$

(a) (3 points)
Complete the following table to describe a possible transition function δ:

	1	␣
q_0		$q_1, 1, R$
q_1		

(b) (2 points)
Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{\_\_\_\_\_\_\_\_}$$

(c) (3 points)
Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\text{␣}) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i, $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \; \text{␣} \; q_1 \; 1 \; R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).

Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e, φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e.

Prove that:

For every partial recursive function g, the set $\{x \in \mathbb{N} \colon g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.

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(a) Every recognizable Language is decidable.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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CSC363 Final Examination

Computational Complexity and Computability   Winter 2023

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Last name

Student ID number

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Q6. Answer each of the following

(a) (3 points)
Give an example (with proof) of a problem in P.

(b) (3 points)
Give three examples of problems which are known to be NP-complete.

Q7. Let A, B, and C be three sets of integers.

Let $D = \{3x \colon x \in A\} \cup \{3x + 1 \colon x \in B\} \cup \{3x + 2 \colon x \in C\}$.

(a) (4 points)
Prove that: $B \leq_p D$.

(b) (2 points)
Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)
Suppose that there exists a set E to which all of A, B, and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k. Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in$ co-NP and $L \notin$ NP, then P $\neq$ NP.

Hint: Prove the contrapositive of the statement.

Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k-*clique* if there exists $C \subseteq V$ such that:

1. C has k-many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) \colon G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)
Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)
Prove that 3SAT $\leq_p L$.

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